Journal of Sports Sciences

ISSN: (Print) (Online) Journal homepage: www.tandfonline.com/journals/rjsp20

Physiological and performance effects of dietary nitrate and N-acetylcysteine supplementation during prolonged heavy-intensity cycling

Rachel Tan, Matthew Black, Joseph Home, Jamie Blackwell, Ida Clark, Lee Wylie, Anni Vanhatalo & Andrew M. Jones

To cite this article: Rachel Tan, Matthew Black, Joseph Home, Jamie Blackwell, Ida Clark, Lee Wylie, Anni Vanhatalo & Andrew M. Jones (2022) Physiological and performance effects of dietary nitrate and N-acetylcysteine supplementation during prolonged heavy-intensity cycling, Journal of Sports Sciences, 40:23, 2585-2594, DOI: 10.1080/02640414.2023.2176052

To link to this article: https://doi.org/10.1080/02640414.2023.2176052

JOURNAL OF SPORTS SCIENCES
2022, VOL. 40, NO. 23, 2585–2594
https://doi.org/10.1080/02640414.2023.2176052

Routledge
Taylor & Francis Group

PHYSIOLOGY AND NUTRITION

OPEN ACCESS Check for updates

Physiological and performance effects of dietary nitrate and N-acetylcysteine supplementation during prolonged heavy-intensity cycling

Rachel Tan, Matthew Black, Joseph Home, Jamie Blackwell, Ida Clark, Lee Wylie, Anni Vanhatalo and Andrew M. Jones

Sport and Health Sciences, College of Life and Environmental Sciences, St. Luke's Campus, University of Exeter, Exeter, UK

ABSTRACT

The purpose of this study was to investigate effects of concurrent and independent administration of dietary nitrate (NO_3^-), administered as NO_3^--rich beetroot juice (BR; ~12.4 mmol of NO_3^-), and N-acetylcysteine (NAC; 70 mg·kg^{-1}) on physiological responses during prolonged exercise and subsequent high-intensity exercise tolerance. Sixteen recreationally active males supplemented with NO_3^--depleted beetroot juice (PL) or BR for 6 days and ingested an acute dose of NAC or maltodextrin (MAL) 1 h prior to performing 1 h of heavy-intensity cycling exercise immediately followed by a severe-intensity time-to-exhaustion (TTE) test in four conditions: 1) PL+MAL, 2) PL+NAC, 3) BR+MAL and 4) BR+NAC. Pre-exercise plasma [NO_3^-] and nitrite ([NO_2^-]) were elevated following BR+NAC and BR+MAL (both $P < 0.01$) compared with PL+NAC and PL+MAL; plasma [cysteine] was increased in PL+NAC and BR+NAC (both $P < 0.01$) compared to PL+MAL. Muscle excitability declined over time during the prolonged cycling bout in all conditions but was better preserved in PL+NAC compared to BR+NAC ($P < 0.01$) and PL+MAL ($P < 0.05$). There was no effect of supplementation on subsequent TTE . These findings indicate that co-ingestion of BR and NAC does not appreciably alter physiological responses during prolonged heavy-intensity cycling or enhance subsequent exercise tolerance.

ARTICLE HISTORY
Received 31 October 2022
Revised 24 January 2023
Accepted 30 January 2023

KEYWORDS
Nitric oxide; reactive oxygen species; endurance; beetroot; antioxidant

Introduction

Dietary nitrate (NO_3^-) supplementation increases nitric oxide (NO) bioavailability and has been reported to enhance muscle contractility (Bailey et al., 2010; Hernández et al., 2012), oxygenation (Ferguson et al., 2013) and exercise performance (Jones et al., 2018). The ergogenic potential of NO_3^- supplementation is more often apparent during intense short-duration exercise (Senefeld et al., 2020) and less often apparent during longer-duration exercise (Christensen et al., 2013; Tan et al., 2018; Wilkerson et al., 2012). Wilkerson et al. (2012) reported that following NO_3^- ingestion, a greater elevation in plasma nitrite (NO_2^-), a precursor for NO production, was associated with greater ergogenic effects during prolonged exercise. Thus, it is possible that the ergogenic potential of NO_3^- supplementation during prolonged exercise requires a more pronounced or protracted elevation of NO bioavailability. While acute dietary NO_3^- supplementation has been shown to result in physiological and ergogenic effects, it has been recommended that combined chronic (several days) and acute (2–3 h pre-exercise) supplementation might optimise efficacy (Jones et al., 2021).

N-acetylcysteine (NAC) donates thiols (i.e., cysteine (CYS), the precursor for glutathione (GSH)), which directly quench reactive oxygen species (ROS) and could protect NO from being scavenged by ROS (Powers & Jackson, 2008). Since ROS production may increase with repetitive skeletal muscle contraction (Reid et al., 1992), such as during prolonged exercise, it is possible that co-ingesting NAC with NO_3^- before prolonged heavy-intensity exercise could preserve NO bioavailability over time due to the antioxidant properties of NAC (Baldelli et al., 2019; Mason et al., 2020). Moreover, neuromuscular fatigue during prolonged exercise is, in part, related to a decline in muscle excitability (Black et al., 2017) which is, in turn, related to Na^+/K^+ ATPase activity. It has been reported that acute NAC supplementation (i.e., 1 h prior to exercise) enhances fatigue resistance via effects on the Na^+/K^+ pump (McKenna et al., 2006) and that NO_3^- attenuates the rise in plasma [K^+] during intense exercise (Wylie et al., 2013b). These results suggest a possible common mechanism of action. Further research is required to assess the independent and combined effects of NO_3^- and NAC supplementation on muscle excitability during prolonged exercise.

The purpose of the present study was, therefore, to investigate the independent and combined effects of NO_3^- and NAC supplementation on NO bioavailability and neuromuscular function during a preload of 1 h of heavy-intensity cycling exercise and on exercise tolerance during a subsequent severe-intensity time-to-exhaustion (TTE) task. We hypothesized that combining NO_3^- and NAC supplementation before 1 h of cycling in the heavy-intensity domain would 1) elevate pre-exercise plasma [NO_3^-], [NO_2^-], [CYS] and [GSH] while reducing the oxidative stress biomarker plasma malondialdehyde ([MDA]), to a greater extent compared to ingestion of NO_3^- or NAC alone; 2) attenuate the expected decline in neuromuscular function to a greater extent than ingestion of NO_3^- or NAC alone; and therefore, 3) improve TTE to a greater extent than ingestion of NO_3^- or NAC alone.

CONTACT Andrew M. Jones a.m.jones@exeter.ac.uk St. Luke's Campus, University of Exeter, EX1 2LU, Exeter, United Kingdom

Methods

Participants

Sixteen recreationally active males (mean ± SD: age 24 ± 6 years, height 1.78 ± 0.05 m, body mass 73 ± 7 kg, VO_{2peak} 52 ± 8 mL $kg^{-1}\cdot min^{-1}$) volunteered to participate in this study. A power calculation based on the effects of NO_3^- (+16%, effect size = 0.78; Bailey et al., 2009) and NAC (+9.6%, effect size = 1.19; unpublished meta-analysis) relative to placebo on TTE for a matched paired t-test indicated a sample size of n = 15 and n = 8, respectively, using a power of 0.80 and alpha of 0.05, and thus, a sample size of n = 16 was chosen. All participants completed a screening and a physical activity readiness questionnaire to ensure that they were safe to exercise and were familiarized to the exercise protocol to minimize any learning effects throughout the study. Participant exclusion criteria included individuals with contraindications to exercise, cardiometabolic disease, users of dietary supplements and smokers (Baranauskas et al., 2022). The protocols, risks and benefits of participating were explained prior to the participants providing written informed consent. This study was approved by the Institutional Research Ethics Committee and conformed to the code of ethics of the Declaration of Helsinki.

Experimental overview

Participants reported to the laboratory on six separate occasions over an ~ 7-week period. All exercise tests were performed on an electrically braked cycle ergometer and controlled using the associated software (Lode Excalibur Sport, Groningen, The Netherlands).

During visit 1, participants completed a ramp incremental test to volitional exhaustion for the determination of gas exchange threshold (GET) and VO_{2peak}. The test began with 3 min of baseline cycling at 20 W, after which the work rate was increased by 30 W/min until task failure. Participants were asked to select a preferred cadence between 70 and 90 rpm and to maintain that cadence throughout the test. Task failure was recorded when the cadence fell by >10 rpm below the selected cadence. The self-selected cadence and seat height and handlebar configuration were recorded for each participant and reproduced on subsequent visits.

For the main experiment, participants commenced baseline cycling at 20 W for 3 min and then completed 1 h of cycling at 20%Δ (where Δ represents the difference between work rates at the GET and VO_{2peak}, plus the work rate at GET) at their self-selected cadence. The TTE commenced immediately after the 1 h was completed at a work rate equivalent to 70%Δ until task failure (Figure 1A.). Blood samples were taken, and neuromuscular function was assessed at baseline and at discrete time points during and post-exercise (see below; Figure 1B.).

Participants were instructed to maintain their habitual physical activity, to avoid consuming foods high in NO_3^- and to refrain from taking dietary supplements or using antibacterial mouthwash as the latter inhibits NO_3^- reduction to NO_2^- (Govoni et al., 2008). In a double-blind, randomized, crossover design, participants were assigned to four conditions to receive NO_3^--rich beetroot juice (BR) or NO_3^--depleted beetroot juice placebo (PL) supplementation for 6 days and an acute dose of NAC or maltodextrin (MAL) 1 h prior to the commencement of the exercise protocol. The dosing protocols we used were based on current recommendations for BR supplementation (A.M. Jones et al., 2021) and previous studies involving oral NAC administration (Corn & Barstow, 2011; Ferreira et al., 2011; Pendyala & Creaven, 1995). Each condition was separated by a wash-out period of 7–10 days (Reid et al., 1994; Wylie et al., 2013a). On day 6 of each of the four supplementation periods, participants reported to the laboratory to complete the experimental protocol. Experimental visits were performed in the morning at the same time of day (8 a.m. ± 2 h). Participants recorded their activity and diet during the 24 h prior to the first experimental visit and were asked to repeat these for subsequent visits. Participants were also instructed to arrive at the laboratory following a 10 h overnight fast, having avoided strenuous exercise and alcohol in the 24 h preceding and caffeine in the 8 h preceding, each experimental visit.

Supplementation

Participants completed four 6-day supplementation periods in which they consumed 2 × 70 mL doses per day of either BR (~6.2 mmol of NO_3^- per 70 mL; Beet It Sport, James White Drinks Ltd., Ipswich, UK) or PL (~0.04 mmol of NO_3^- per 70 mL; Beet It Sport, James White Drinks Ltd., Ipswich, UK). On day 1–5 of each supplementation period, participants consumed one 70 mL beverage in the morning and one in the evening. On the experimental day, participants consumed 2 × 70 mL of their allocated beverage 2.5 h prior to exercise (Wylie et al., 2013a) and ingested 70 mg kg^{-1} of NAC (Piping Rock Health Products, New York, USA, 600 mg NAC per capsule; Ferreira et al., 2011) or maltodextrin (MAL: Bulk Powders, Sports Supplements Ltd., Essex, UK, 600 mg per capsule) 1 h prior to exercise (Corn & Barstow, 2011) before arriving to the laboratory. Capsules for NAC and MAL were identical in size and appearance. Peppermint-scented oil was applied to the outside of all capsules to mask the sulphur smell of NAC.

The four supplementation conditions were (1) PL with MAL (PL+MAL), (2) PL with NAC (PL+NAC), (3) BR with MAL (BR+MAL) and (4) BR with NAC (BR+NAC). Participants were contacted on a daily basis to ensure the consumption of supplement via text or email.

Blood analyses

Venous blood was sampled at baseline, 30 min and 60 min during the 1 h cycling bout and within 15 s post-TTE. Blood samples were obtained from a cannula (Insyte-W™ Becton-Dickinson, Madrid, Spain) that was inserted in the participant's antecubital vein and drawn into 6 mL lithium heparin tubes (Vacutainer, Becton-Dickinson, New Jersey, USA). For plasma [CYS] and [GSH] analysis (Newton et al., 1981; Shen et al., 2015), within 1 min of collection, 200 µL of whole blood was added to 800 µL of ice-cold deionized water and centrifuged at 16000G, 4°C for 10 min to liberate and detect biological thiols through erythrocyte lysation. Following this, 60 µL of the supernatant was extracted and added to 140 µL of reaction buffer (Tris-HCl, 100 mM, pH 9.5, 0.1 mM Diethylenetriaminepentaacetic acid, DPTA), to release and stabilize thiols, and 100 µL of 10 mM

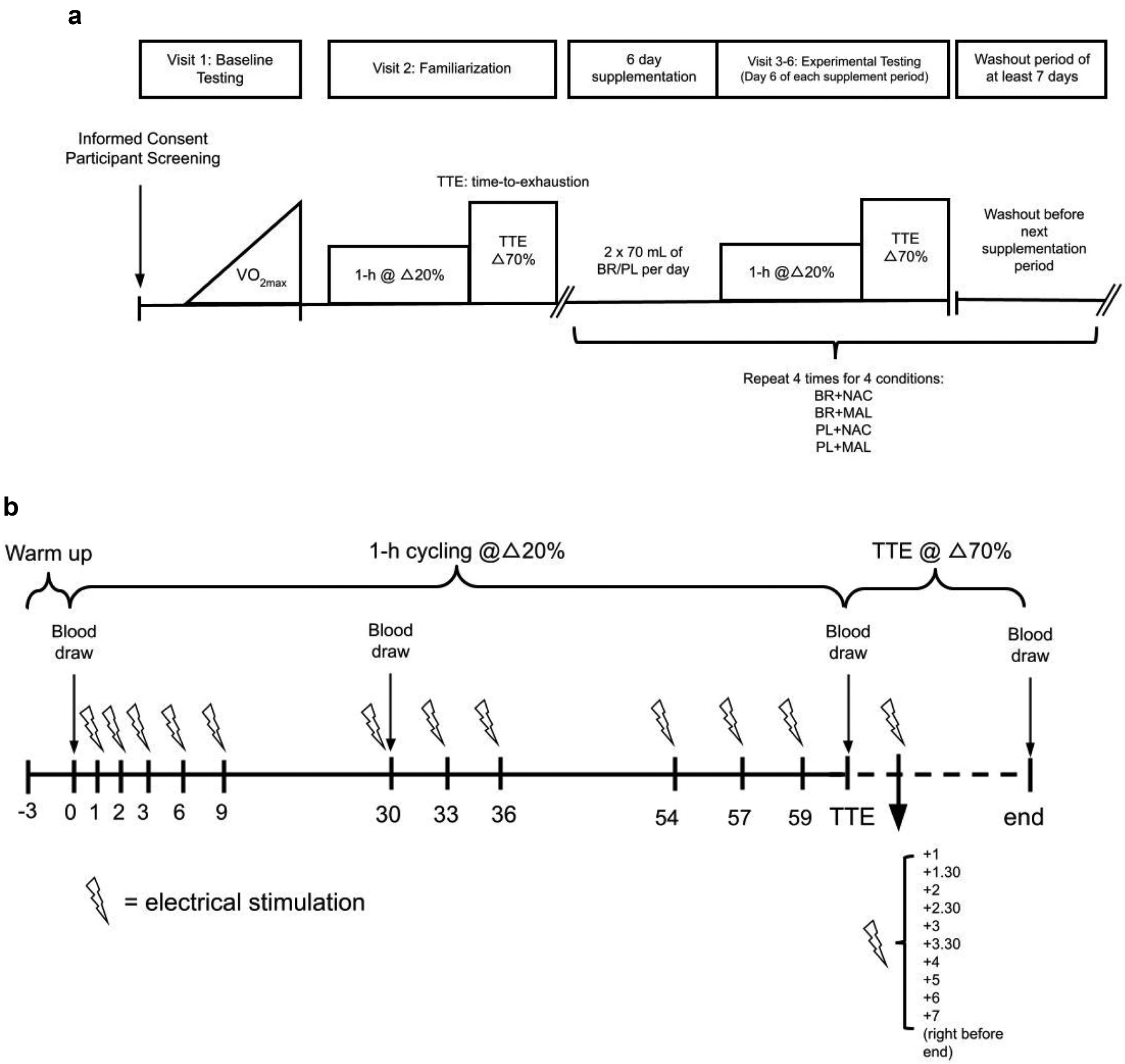

Figure 1. Schematic diagram of the experimental protocol.

monobromobimane in acetonitrile (CH_3CN) was added (Newton et al., 1981; Shen et al., 2015). After 30 min of incubation in the dark, 100 µL of ice-cold 200 mM sulfosalicyclic acid solution was added. The mixture was then incubated on ice for 10 min and then centrifuged at16000G at 4°C for 10 min. The supernatant was extracted and frozen at −80°C for subsequent injection into the high performance liquid chromatography (HPLC) system (Perkin Elmer® Flexar™ Chomera® software version v.4.1.2.6410, Perkin Elmer Inc., MA, USA). The supernatant was diluted (1:20 fold) in deionized water and injected (10 µL) into the HPLC for analysis. The HPLC mobile phase was 0.1% trifluoroacetic acid aqueous and 0.1% trifluoroacetic acid in acetonitrile running through a 250 × 4.6 C18 5 µm Brownlee validated column (Perkin Elmer Inc., MA, USA) at 0.9 mL/min flow rate, with fluorescence detection at 390 nm excitation and 475 nm emission, determined against standard curves from commercially available CYS and GSH (Figure 2).

The remaining whole blood samples were centrifuged within 2 min of collection at 4000 rpm and 4°C for 10 min, and plasma was then extracted and frozen at −80°C. For the analysis of plasma [NO_3^-] and [NO_2^-], samples were deproteinized and analysed via gas phase chemiluminescence as previously described (Tan et al., 2018). For plasma MDA analysis (Moselhy et al., 2013), 100 µL of plasma was added to 25 µL of 3 M NaOH and heated at 60°C in a water bath for 30 min for alkaline hydrolysis (Grotto et al., 2007). Following this, 1 mL of 0.05 sulphuric acid and 500 µL of 20% w/v trichloroacetic acid were added. The samples were centrifuged at 3000 rpm for 10 min, and 1 mL of the supernatant was extracted into a new heat-

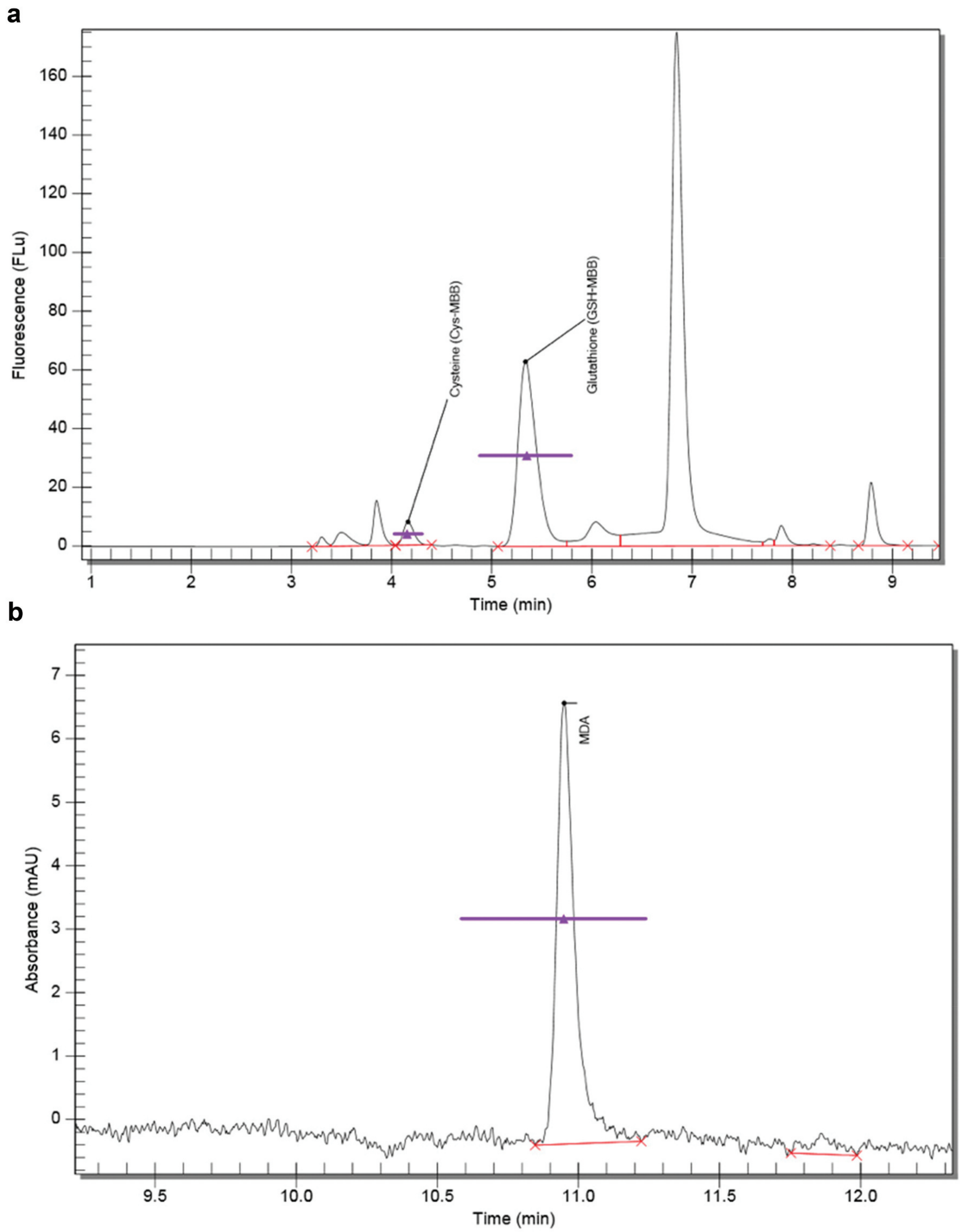

Figure 2. Individual representative chromatogram/trace for plasma cysteine and glutathione (panel A) and plasma malondialdehyde (panel B) assessed via high performance liquid chromatography.

resistant tube. Following this, 500 µL of 0.355% w/v thiobarbituric acid was added and the samples were vortexed. Samples were heated at 90°C for 40 min. Once cooled, the samples were centrifuged at 3000 rpm for 10 min and were injected (25 µL) into the HPLC for analysis.

The HPLC mobile phase was 0.1% formic acid aqueous solution and methanol organic running through a 150 × 4.6 C18 5 µm Brownlee validated column (Perkin Elmer Inc., MA, USA) at a 1.0 mL/min flow rate, with photodiode array detection analytical absorbance measured at 532 nm, reference 395 nm, determined against standard curves from commercially available MDA (Figure 2).

Neuromuscular function

Neuromuscular function was assessed at baseline, 0 min, 30 min, 57 min and post-TTE as previously described (Black et al., 2017). Briefly, EMG was recorded continuously on the *vastus lateralis* (VL) using electrodes (DE2.1; Delsys, Boston, MA) positioned over the muscle belly after shaving and cleaning the skin [surface EMG for non-invasive measurement of muscles (SENIAM) guidelines]. The ground electrode was positioned on the patella. EMG signals were pre-amplified (1,000x), band-pass filtered (20–450 Hz, Bagnioli-8; Delsys), and digitized at a sampling rate of 2,000 Hz and a resolution of 16 bits using a Power 1401 mk-II analog-to-digital converter and Spike 2 data collection software run by custom-written sampling configuration (CED; Cambridge Electronic Design, UK). For transcutaneous femoral nerve stimulation, an adhesive cathode (Boots UK, Nottingham, England) was placed ~2 cm medial of the femoral pulse, and an adhesive anode (Boots UK) was placed at the anterior aspect of the iliac crest. Single electrical pulses were generated by a stimulator (DS7 A; Digitimer, UK) and randomly delivered at an identified crank angle for all trials for each participant (65 ± 5° relative to the top dead centre; Sidhu et al., 2012).

A standard M-wave recruitment curve protocol was completed during each laboratory visit. Participants cycled at 20 W below GET throughout the recruitment curve protocol. A single-pulse electrical stimulation (200 μs) was delivered at the individually identified crank angle as described above. The current was increased in 20-mA increments until the M-wave amplitude reached a plateau at the maximal M-wave amplitude (M_{max}). A pulse of 120% M_{max} current was applied during the exercise tests (mean stimulation intensity, 290 ± 48 mA).

EMG signals from the VL were processed using a custom-written script to measure peak-to-peak M-wave amplitude and M-wave area. The root mean square (RMS) of the EMG signal (an index of the power of the signal) was calculated as the mean over a 25-ms pre-stimulation period at each stimulation time point. The EMG RMS amplitudes were normalized to the corresponding values attained after 1 min of exercise during each trial to evaluate temporal changes in the voluntary muscle activation level (i.e., EMG RMS amplitude). M-wave parameters were normalized to baseline values to evaluate temporal changes in the peripheral neuromuscular excitability (i.e., M-wave amplitude and area). Voluntary EMG RMS amplitude was normalized to the M-wave amplitude to assess changes in neural drive (RMS/M; Millet & Lepers, 2004). The rates of change in M-wave and EMG parameters from baseline to post-TTE were calculated for each exercise to quantify the rate of neuromuscular fatigue development. The coefficients of variation (CV%) for M-wave amplitude and M-wave total area between trials were 10% and 8%, respectively, and between stimulations, they were 11% and 9%, respectively.

Statistical analyses

Two-way (condition x time) repeated ANOVAs were used to analyse differences in plasma [NO_3^-], [NO_2^-], [CYS], [GSH], [MDA] and neuromuscular function during the 1 h heavy-intensity exercise bout. One-way repeated measures ANOVAs were used to analyse differences in TTE and physiological performance differences in the mean and change values from pre- to post-1 h exercise and post-TTE. Significant main and interaction effects were further explored using Fisher's LSD test. A full set of blood data were only available for n = 14 for time points 0 to 60 min and n = 13 for time point post-TTE due to problems with cannulation and/or blood draw. Neuromuscular data were analysed for a subset of n = 12. Data for TTE are reported for n = 15 owing to n = 1 having foot slip out of pedal. Statistical significance was accepted at $P \leq 0.05$. Results are presented as mean ± SD unless otherwise stated.

Results

Plasma [NO_3^-] and [NO_2^-]

There were an interaction (condition x time) ($P < 0.05$) and effect of condition ($P < 0.01$), but no effect of time ($P > 0.05$) for plasma [NO_3^-] (Figure 3A). Plasma [NO_3^-] was greater in BR +NAC ($P < 0.01$) and BR+MAL ($P < 0.01$) compared to PL+NAC and PL+MAL at all time points. Plasma [NO_3^-] in BR+NAC compared to BR+MAL was elevated at 60 min ($P < 0.05$). There was no difference in plasma [NO_3^-] between PL+NAC and PL+MAL at any time point ($P > 0.05$). Plasma [NO_3^-] declined from 0 min to 60 min in BR+MAL ($P < 0.05$), PL+MAL ($P < 0.05$) and PL+NAC ($P < 0.01$) but not in BR+NAC ($P > 0.05$). Plasma [NO_3^-] did not change from 60 min to post-TTE in any condition ($P > 0.05$)

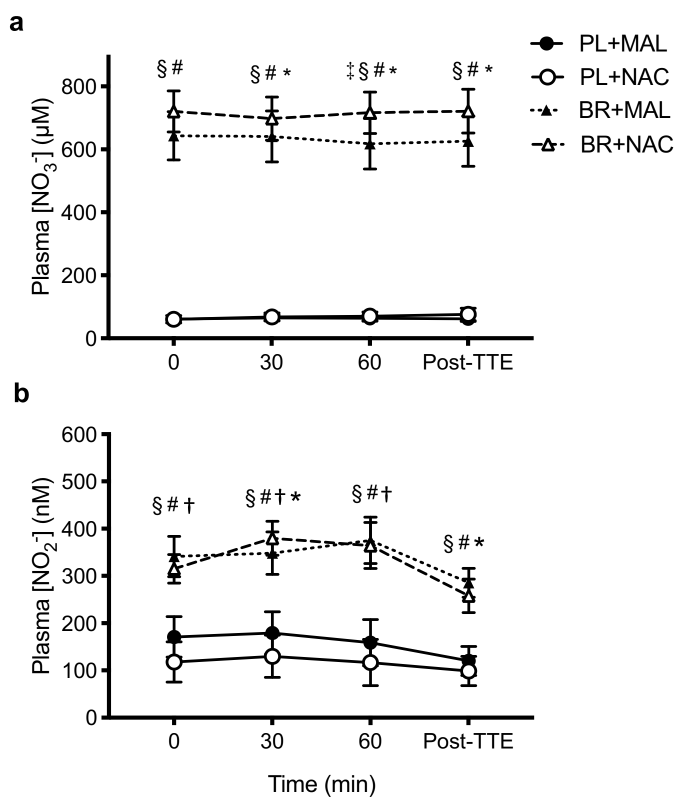

Figure 3. Mean ± SE plasma nitrate (NO_3^-; panel A) and nitrite (NO_2^-; panel B) over 1 h of heavy-intensity exercise and a subsequent TTE following: PL+MAL (solid circle and solid line); PL+NAC (open circle and solid line); BR+MAL (solid triangle and dotted line); and BR+NAC (open triangle and dotted line). * = significantly different from 0 min ($P < 0.05$), † = significantly different from TTE ($P < 0.05$), § = significantly different in BR+NAC compared to PL+MAL and PL +NAC ($P < 0.01$), # = significantly different in BR+MAL compared to PL+MAL and PL+NAC ($P < 0.01$), ‡ = significantly different in BR+NAC compared to BR+MAL ($P < 0.05$). Data shown for a subset of n = 14 from 0 to 60 min and for n = 13 for post-TTE.

There were an interaction (condition x time) ($P = 0.05$), effect of time ($P < 0.01$), and effect of condition ($P < 0.01$) for plasma [NO_2^-] (Figure 3B). At baseline, plasma [NO_2^-] was elevated in BR+NAC ($P < 0.01$) and BR+MAL ($P < 0.01$) compared to PL+NAC and PL+MAL. There was no difference in plasma [NO_2^-] at baseline between BR+NAC and BR+MAL ($P > 0.05$) or PL+NAC and PL+MAL ($P > 0.05$). Plasma [NO_2^-] did not change over 1 h of exercise in any of the conditions ($P > 0.05$). Plasma [NO_2^-] declined from 60 min to post-TTE in BR+NAC ($P < 0.05$), BR +MAL ($P < 0.05$) and PL+NAC ($P < 0.05$) but not PL+MAL ($P > 0.05$).

Plasma [CYS], [GSH] and [MDA]

There was an effect of time ($P < 0.01$) for plasma [CYS] (Figure 4A). Plasma [CYS] was greater in PL+NAC (11 ± 5 μM, $P < 0.01$) and BR+NAC (15 ± 6 μM, $P < 0.01$) compared to PL+MAL (7 ± 3 μM) at rest. Plasma [CYS] increased over time compared to 0 min ($P < 0.01$). There was a difference between the change in plasma [CYS] from 0 to 60 min in BR+NAC and BR+MAL compared to PL+NAC and PL+MAL ($P < 0.05$; Figure 4B). Plasma [GSH] increased from 0 to 60 min ($P < 0.01$) and post-TTE ($P < 0.01$; Figure 4C). There was no effect of condition in the change of plasma [GSH] from 0 to 60 min ($P > 0.05$; Figure 4D). There were no interaction, effect of time or effect of condition on plasma [MDA] (all $P > 0.05$; Figure 4E) or the change in plasma [MDA] from 0 to 60 min (all $P > 0.05$; Fig. 4F).

Peripheral neuromuscular excitability: M-wave amplitude and M-wave area

There was an effect of time ($P < 0.001$) but no effect of condition on the M-wave amplitude ($P > 0.05$) or M-wave area ($P > 0.05$, Figure 5A and B). There was an effect of condition ($P < 0.05$) in the change of M-wave amplitude

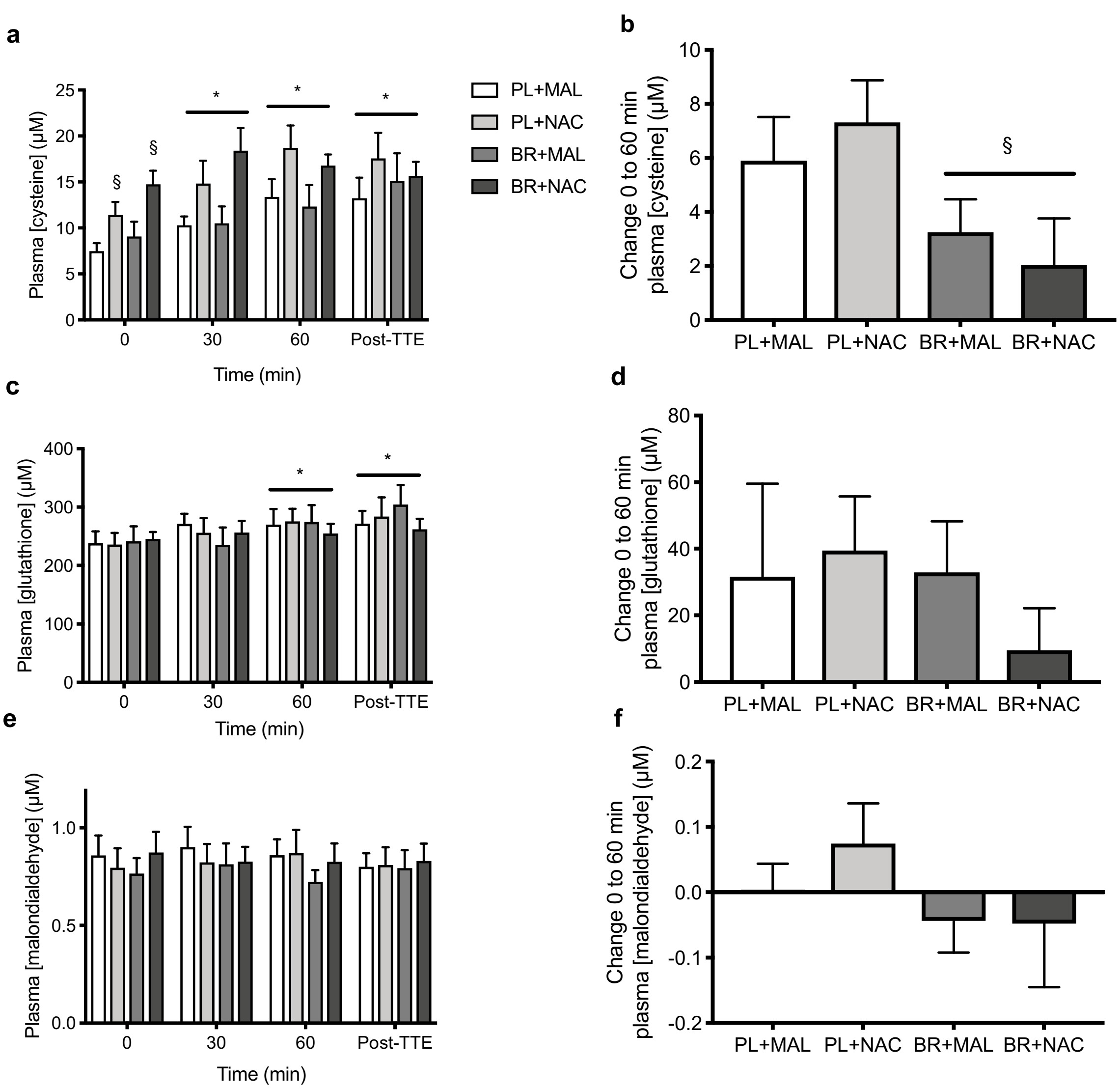

Figure 4. Mean ± SE plasma cysteine (panel A), glutathione (panel C) and malondialdehyde (panel E) over 1 h of heavy-intensity exercise and a subsequent TTE. Change from 0 to 60 min in plasma [cysteine] (panel B), change in plasma [glutathione] (panel D) and change in plasma [MDA] (panel F) following four supplementation conditions of PL+MAL, PL+NAC, BR+MAL and BR+NAC. * = significantly different from 0 min ($P < 0.01$). § = significantly different to PL+NAC and PL+MAL ($P < 0.05$). Data shown for a subset of n = 14 for time points 0 to 60 min and n = 13 for time point post-TTE for plasma [cysteine] and [glutathione] and a subset of n = 13 for plasma [malondialdehyde].

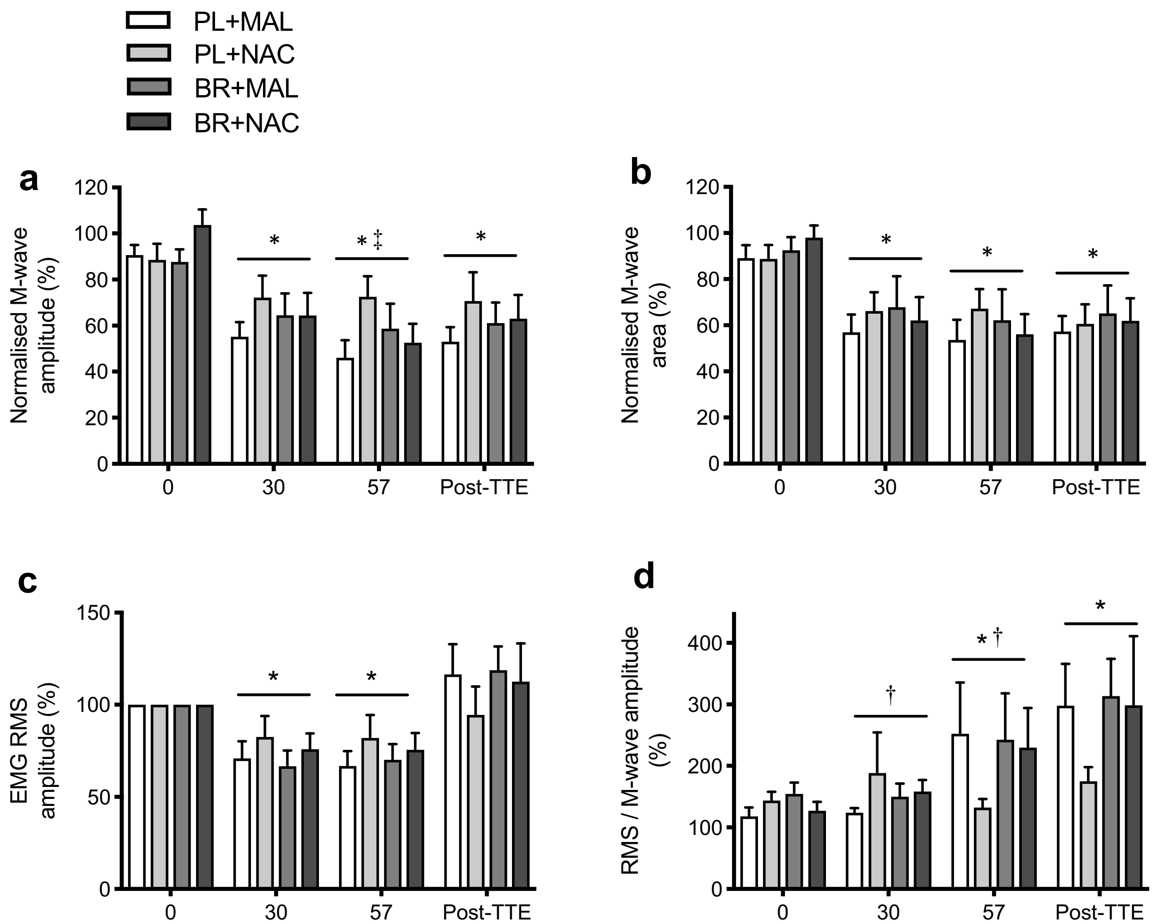

Figure 5. Compound muscle action potential (M-wave) amplitude and M-wave area (normalized to maximum M-wave during baseline pedalling) indicating peripheral neuromuscular excitability (panel A and B); voluntary electromyographic root mean square (EMG RMS) amplitude (normalized to M-wave amplitude at 1 min of exercise) indicating muscle activation level (panel C) and RMS/M-wave (normalized to corresponding M-wave amplitude at each measurement time point) (panel D) during 1 h of heavy-intensity cycling at 0, 30 and 57 min, as well as at the end of the time-to-exhaustion (Post-TTE), following four supplementation conditions of PL +MAL, PL+NAC, BR+MAL and BR+NAC. * = significantly different from 0 min ($P < 0.01$); ‡ = significantly different from 30 min; † = significantly different from post-TTE ($P < 0.01$). Data shown for the subset of $n = 12$.

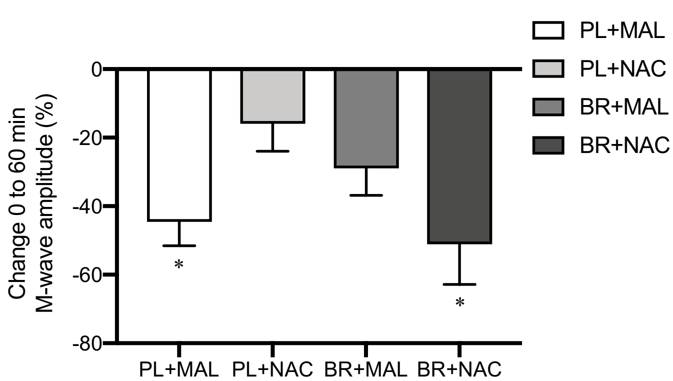

Figure 6. Change from 0 to 60 min in compound action potential (M-wave) amplitude indicating peripheral neuromuscular excitability, following four supplementation conditions of PL+MAL, PL+NAC, BR+MAL and BR+NAC. * = significantly different to PL+NAC. Data shown for subset of $n = 12$.

from 0 to 57 min (Figure 6), with the change being smaller in PL+NAC (−16 ± 28%) compared to PL+MAL (−45 ± 24%; $P < 0.05$) and BR+NAC (−51 ± 41%; $P < 0.01$). There was no effect of condition ($P > 0.05$) or time ($P > 0.05$) on the change in the M-wave area from 0 to 57 min.

Voluntary activation and neural drive

Voluntary muscle activation level (EMG RMS amplitude) and neural drive (RMS/M-wave amplitude) at 0, 30 and 57 min and post-TTE are shown in Figure 5C and D. There was an effect of time ($P < 0.001$) but no effect of condition on EMG RMS ($P > 0.05$) or RMS/M-wave amplitude ($P > 0.05$). There was no effect of condition ($P > 0.05$) or time ($P > 0.05$) on the change from 0 to 57 min in voluntary activation or neural drive.

Time-to-exhaustion

No differences were found between conditions for TTE at 70%Δ (PL+MAL: 246 ± 134 vs. PL+NAC: 238 ± 149 vs. BR+MAL: 252 ± 118 vs. BR+NAC: 235 ± 140 s; $P > 0.05$).

Discussion

This study investigated the independent and combined influence of BR and NAC administration on NO bioavailability and neuromuscular function during a preload of 1 h of heavy-intensity cycling and on subsequent severe-intensity exercise tolerance in humans. The major novel findings were that 1) co-ingestion of BR and NAC (BR+NAC) better maintained plasma [NO_3^-] during exercise; 2) the decline in muscle excitability during the 1 h exercise bout was significantly attenuated in NAC alone (PL+NAC) compared to PL+MAL and BR+NAC and 3) there was no influence of any supplementation condition on severe-intensity exercise tolerance. These findings are partly in agreement with our first hypothesis that BR+NAC would better preserve plasma [NO_3^-] during prolonged exercise although this did not translate into better preserved plasma [NO_2^-]. However, these findings do not support our second and third hypotheses as neither independent nor combined ingestion of BR and NAC were effective at improving muscle excitability or TTE.

Effects of BR and NAC supplementation on blood biomarkers

Plasma [NO_3^-] was better preserved in BR+NAC compared to BR +MAL at 1 h, suggesting that co-ingestion of BR and NAC better maintains plasma [NO_3^-] during prolonged exercise. However, there was no additional increase in plasma [NO_2^-] in BR+NAC compared to BR+MAL, and thus, it is unclear whether co-ingesting BR and NAC appreciably augmented NO bioavailability (Wylie et al., 2013a). The failure of BR+NAC to augment plasma [NO_2^-] compared to BR alone is consistent with the limited existing data (Bailey et al., 2011; Crum et al., 2018). One possible explanation could be a limitation in the rate of reduction of NO_3^- to NO_2^- by the oral microbiota (Govoni et al., 2008) or differences in the rate of entry of NO_3^- or NO_2^- into skeletal muscle (Piknova et al., 2015). It is also possible that the exercise task did not appreciably increase ROS, thereby negating any antioxidant effects of NAC in preserving NO, a possibility that is supported by the absence of changes to plasma [MDA]. It should also be noted that BR contains numerous antioxidant compounds (Wootton-Beard & Ryan, 2012) such that NAC may not have been required to quench ROS and maintain NO bioavailability.

Following BR+NAC and PL+NAC, plasma [CYS] increased, which is consistent with other reports (Ferreira et al., 2011; Medved et al., 2004), but plasma [GSH] and [MDA] did not change. Given that CYS is the precursor to GSH, an increase in plasma [GSH] might be expected following NAC supplementation. It is possible that methodological differences between studies, such as analysis of whole blood vs. plasma, administration of capsules vs. solution and assessment of total vs. reduced/oxidized GSH, may impact the ability to observe changes in GSH (Corn & Barstow, 2011; Ferreira et al., 2011). Despite this, it is worth mentioning that previous investigations have observed favourable performance effects following NAC supplementation independent of changes to plasma [GSH] (Kelly et al., 2009). Plasma [MDA] was our only marker of oxidative damage via measuring lipid peroxidation (Janero, 1990), and the lack of change in plasma [MDA] could suggest that the exercise protocol did not induce significant changes to lipid peroxidation or that GSH promoted a more favourable cellular environment. However, it should be recognised that MDA and GSH provide only limited information on overall oxidative stress and redox balance, and the inclusion of additional biomarkers would have provided greater insight (Cobley et al., 2017). It is also possible that NAC supplementation is only effective in individuals with an antioxidant deficiency (Margaritelis et al., 2020) and this might be one explanation for the limited effects of NAC in the present study.

Effects of BR and NAC supplementation on neuromuscular function

Muscle excitability (M-wave amplitude and area) and voluntary activation (EMG RMS) declined over the 1 h exercise bout, irrespective of supplementation, which is consistent with previous findings (Black et al., 2017). Interestingly, muscle excitability was better preserved in PL+NAC compared to PL+MAL and BR+NAC, results which are partly consistent with earlier studies (McKenna et al., 2006; Medved et al., 2004). The mechanisms responsible for improved muscle excitability following independent ingestion of NAC could be related to preserved membrane depolarization via enhanced Na^+/K^+ ATPase activity (Gandevia, 2001; McKenna et al., 2006) and thus better regulation of muscle and plasma [K^+] (Medved et al., 2004). We observed that BR ingestion did not impact muscle excitability despite a previous report that plasma [K^+] was reduced during intense exercise following BR compared to placebo (Wylie et al., 2013b). Very few studies have examined the effect of BR on neuromuscular function during voluntary contractions. One study reported that motor unit firing rate was enhanced following BR (Esen et al., 2022) but this was not confirmed in other studies (Husmann et al., 2019; Thurston et al., 2021). When BR and NAC are consumed together, it is possible that antagonistic molecular signalling occurs, negating any possible positive effects when either supplement is ingested alone (Powers & Jackson, 2008). Further research is required to resolve the potential effects of independently ingesting BR and co-ingesting BR+NAC on neuromuscular function during prolonged exercise.

Effects of BR and NAC supplementation on exercise tolerance

TTE during severe-intensity exercise was not different between conditions. This is in contrast to previous reports that BR (Bailey et al., 2010, 2009) and NAC (McKenna et al., 2006; Medved et al., 2004) supplementation may independently improve performance during TTE tests. It has been suggested that BR is more likely to enhance performance in less trained individuals (Senefeld et al., 2020), whereas NAC is more likely to enhance performance in well-trained individuals (Medved et al., 2004) and, therefore, combining these supplements might be ineffective. A limitation of this study is that it only included recreationally active males, and further research is required to examine other populations. Additionally, our exercise protocol was somewhat unusual in that it involved a prolonged (1 h) heavy-intensity pre-load prior to the TTE test. The available evidence suggests that BR might be more beneficial during shorter-duration, higher-intensity activities (Senefeld et al., 2020) and it is possible that the performance of the pre-load reduces the potential efficacy of BR supplementation (Wilkerson et al., 2012). Consistent with this, two previous studies that employed a prolonged bout of exercise prior to the exercise performance task also failed to find a positive effect of BR (Christensen et al., 2013; Tan et al., 2018). In contrast, NAC supplementation appears to favour exercise performance following a longer duration pre-load (≥ 45 min; McKenna et al., 2006; Medved et al., 2004; cf., Bailey et al., 2011). Thus, because BR and NAC are more likely to be ergogenic in different scenarios, combining them may not be efficacious. Also, under the conditions of the present study in which both BR and NAC had limited effects when taken in isolation, it might not be surprising to find a lack of synergy when they are combined.

Conclusion

Combining BR and NAC better maintained the elevation of plasma [NO_3^-] during 1 h of heavy-intensity cycling compared to BR alone; however, there was no synergistic effect of co-ingesting BR and NAC on maintaining plasma [NO_2^-] compared to BR alone. Muscle excitability during 1 h of heavy-intensity cycling was better preserved following independent ingestion of NAC, but there was no effect of ingestion of BR alone or co-ingestion of BR and NAC on muscle excitability compared to placebo. Moreover, TTE during severe-intensity exercise was not different between the four conditions such that, under the conditions of the present study, BR and NAC did not enhance exercise performance either when taken separately or when combined.

Acknowledgments

We thank Adam Linoby, Felix Leonard-Fox, Tuesday Platt and James Lewis for assistance with data collection and data analysis. We thank Dr Stephen Bailey for discussions regarding the study design.

Disclosure statement

The authors report there are no competing interests to declare.

Funding

This research received no external funding.

Author Contributions

R.T., A.V. and A.M.J. contributed to conceptualization; R.T., M.I.B., J.H., J.R.B. and I.C. contributed to methodology; R.T., M.I.B. and J.R.B. contributed to formal analysis; R.T., J.H. and I.C. contributed to investigation; R.T., M.I.B. contributed to data curation; R.T. contributed to writing – original draft preparation; R.T., M.I.B., A.V. and A.M.J. contributed to writing – review and editing L.J.W., A.V. and A.M.J. contributed to supervision; R.T. contributed to project administration. All authors have read and agreed to the published version of the manuscript.

References

Bailey, S. J., Fulford, J., Vanhatalo, A., Winyard, P. G., Blackwell, J. R., DiMenna, F. J., Wilkerson, D. P., Benjamin, N., & Jones, A. M. (2010). Dietary nitrate supplementation enhances muscle contractile efficiency during knee-extensor exercise in humans. *Journal of Applied Physiology (Bethesda, Md.: 1985)*, *109*(1), 135–148. https://doi.org/10.1152/japplphysiol.00046.2010

Bailey, S. J., Winyard, P. G., Blackwell, J. R., Vanhatalo, A., Lansley, K. E., Dimenna, F. J., Wilkerson, D. P., Campbell, I. T., & Jones, A. M. (2011). Influence of N-acetylcysteine administration on pulmonary O2 uptake kinetics and exercise tolerance in humans. *Respiratory Physiology & Neurobiology*, *175*(1), 121–129. https://doi.org/10.1016/j.resp.2010.10.002

Bailey, S. J., Winyard, P., Vanhatalo, A., Blackwell, J. R., Dimenna, F. J., Wilkerson, D. P., Tarr, J., Benjamin, N., & Jones, A. M. (2009). Dietary nitrate supplementation reduces the O2 cost of low-intensity exercise and enhances tolerance to high-intensity exercise in humans. *Journal of Applied Physiology (Bethesda, Md.: 1985)*, *107*(4), 1144–1155. https://doi.org/10.1152/japplphysiol.00722.2009

Baldelli, S., Ciccarone, F., Limongi, D., Checconi, P., Palamara, A. T., & Ciriolo, M. R. (2019). Glutathione and Nitric Oxide: Key Team Players in Use and Disuse of Skeletal Muscle. *Nutrients*, *11*(10), 2318. https://doi.org/10.3390/nu11102318

Baranauskas, M. N., Freemas, J. A., Tan, R., & Carter, S. J. (2022). Moving beyond inclusion: Methodological considerations for the menstrual cycle and menopause in research evaluating effects of dietary nitrate on vascular function. *Nitric Oxide: Biology and Chemistry*, *118*, 39–48. https://doi.org/10.1016/j.niox.2021.11.001

Black, M. I., Jones, A. M., Blackwell, J. R., Bailey, S. J., Wylie, L. J., McDonagh, S. T., Thompson, C., Kelly, J., Sumners, P., Mileva, K. N., Bowtell, J. L., & Vanhatalo, A. (2017). Muscle metabolic and neuromuscular determinants of fatigue during cycling in different exercise intensity domains. *Journal of Applied Physiology (Bethesda, Md.: 1985)*, *122*(3), 446–459. https://doi.org/10.1152/japplphysiol.00942.2016

Christensen, P. M., Nyberg, M., & Bangsbo, J. (2013). Influence of nitrate supplementation on VO_2 kinetics and endurance of elite cyclists. *Scandinavian Journal of Medicine in Science & Sports*, *23*(1), e21–e31. https://pubmed.ncbi.nlm.nih.gov/23020760/

Cobley, J.N., Close, G.L., Bailey, D.M., & Davison, G.W. (2017). Exercise redox biochemistry: Conceptual, methodological and technical recommendations. *Redox Biology*, *12*, 540–548. https://doi.org/10.1016/j.redox.2017.03.022

Corn, S. D., & Barstow, T. J. (2011). Effects of oral N-acetylcysteine on fatigue, critical power, and W′ in exercising humans. *Respiratory Physiology & Neurobiology*, *178*(2), 261–268. https://doi.org/10.1016/j.resp.2011.06.020

Crum, E. M., Barnes, M. J., & Stannard, S. R. (2018). Multiday pomegranate extract supplementation decreases oxygen uptake during submaximal cycling exercise, but cosupplementation with N-acetylcysteine negates the effect. *International Journal of Sport Nutrition and Exercise Metabolism*, *28*(6), 586–592. https://doi.org/10.1123/ijsnem.2017-0407

Esen, O., Faisal, A., Zambolin, F., Bailey, S. J., & Callaghan, M. J. (2022). Effect of nitrate supplementation on skeletal muscle motor unit activity during isometric blood flow restriction exercise. *European Journal of Applied Physiology*, *122*(7), 1683–1693. https://doi.org/10.1007/s00421-022-04946-y

Ferguson, S. K., Hirai, D. M., Copp, S. W., Holdsworth, C. T., Allen, J. D., Jones, A. M., Musch, T. I., & Poole, D. C. (2013). Impact of dietary nitrate supplementation via beetroot juice on exercising muscle vascular control in rats. *The Journal of Physiology*, *591*(2), 547–557. https://doi.org/10.1113/jphysiol.2012.243121

Ferreira, L. F., Campbell, K. S., & Reid, M. B. (2011). N-acetylcysteine in handgrip exercise: Plasma thiols and adverse reactions. *International Journal of Sport Nutrition and Exercise Metabolism*, *21*(2), 146–154. https://doi.org/10.1123/ijsnem.21.2.146

Gandevia, S. C. (2001). Spinal and supraspinal factors in human muscle fatigue. *Physiological Reviews*, *81*(4), 1725–1789. https://doi.org/10.1152/physrev.2001.81.4.1725

Govoni, M., Jansson, E. A., Weitzberg, E., & Lundberg, J. O. (2008). The increase in plasma nitrite after a dietary nitrate load is markedly attenuated by an antibacterial mouthwash. *Nitric Oxide: Biology and Chemistry*, *19*(4), 333–337. https://doi.org/10.1016/j.niox.2008.08.003

Grotto, D., Santa Maria, L. D., Boeira, S., Valentini, J., Charao, M. F., Moro, A. M., Nascimento, P. C., Pomblum, V. J., & Garcia, S. C. (2007). Rapid quantification of malondialdehyde in plasma by high performance liquid chromatography-visible detection. *Journal of*

Pharmaceutical and Biomedical Analysis, *43*(2), 619–624. https://doi.org/10.1016/j.jpba.2006.07.030

Hernández, A., Schiffer, T. A., Ivarsson, N., Cheng, A. J., Bruton, J. D., Lundberg, J. O., Weitzberg, E., & Westerblad, H. (2012). Dietary nitrate increases tetanic [Ca2+]i and contractile force in mouse fast-twitch muscle. *The Journal of Physiology*, *590*(15), 3575–3583. https://doi.org/10.1113/jphysiol.2012.232777

Husmann, F., Bruhn, S., Mittlmeier, T., Zschorlich, V., & Behrens, M. (2019). Dietary nitrate supplementation improves exercise tolerance by reducing muscle fatigue and perceptual responses. *Frontiers in Physiology*, *10*, 404. https://doi.org/10.3389/fphys.2019.00404

Janero, D. R. (1990). Malondialdehyde and thiobarbituric acid-reactivity as diagnostic indices of lipid peroxidation and peroxidative tissue injury. *Free Radical Biology & Medicine*, *9*(6), 515–540. https://doi.org/10.1016/0891-5849(90)

Jones, A. M., Thompson, C., Wylie, L. J., & Vanhatalo, A. (2018). Dietary Nitrate and Physical Performance. *Annual Review of Nutrition*, *38*, 303–328. https://doi.org/10.1146/annurev-nutr-082117-051622

Jones, A. M., Vanhatalo, A., Seals, D. R., Rossman, M. J., Piknova, B., & Jonvik, K. L. (2021). Dietary nitrate and nitric oxide metabolism: mouth, circulation, skeletal muscle, and exercise performance. *Medicine & Science in Sports & Exercise*, *53*(2), 280–294. https://doi.org/10.1249/MSS.0000000000002470

Kelly, M. K., Wicker, R. J., Barstow, T. J., & Harms, C. A. (2009). Effects of N-acetylcysteine on respiratory muscle fatigue during heavy exercise. *Respiratory Physiology & Neurobiology*, *165*(1), 67–72. https://doi.org/10.1016/j.resp.2008.10.008

Margaritelis, N. V., Paschalis, V., Theodorou, A. A., Kyparos, A., & Nikolaidis, M. G. (2020). Antioxidant supplementation, redox deficiencies and exercise performance: A falsification design. *Free Radical Biology and Medicine*, *158*, 44–52. https://doi.org/10.1016/j.freeradbiomed.2020.06.029

Mason, S. A., Trewin, A. J., Parker, L., & Wadley, G. D. (2020). Antioxidant supplements and endurance exercise: Current evidence and mechanistic insights. *Redox Biology*, *35*, 101471. https://doi.org/10.1016/j.redox.2020.101471

McKenna, M. J., Medved, I., Goodman, C. A., Brown, M. J., Bjorksten, A. R., Murphy, K. T., Petersen, A. C., Sostaric, S., & Gong, X. (2006). N-acetylcysteine attenuates the decline in muscle Na+,K+-pump activity and delays fatigue during prolonged exercise in humans. *The Journal of Physiology*, *576*(Pt 1), 279–288. https://doi.org/10.1113/jphysiol.2006.115352

Medved, I., Brown, M. J., Bjorksten, A. R., Murphy, K. T., Petersen, A. C., Sostaric, S., Gong, X., & McKenna, M. J. (2004). N-acetylcysteine enhances muscle cysteine and glutathione availability and attenuates fatigue during prolonged exercise in endurance-trained individuals. *Journal of Applied Physiology (Bethesda, Md.: 1985)*, *97*(4), 1477–1485. https://doi.org/10.1152/japplphysiol.00371.2004

Millet, G. Y., & Lepers, R. (2004). Alterations of neuromuscular function after prolonged running, cycling and skiing exercises. *Sports Medicine*, *34*(2), 105–116. https://doi.org/10.2165/00007256-200434020-00004

Moselhy, H. F., Reid, R. G., Yousef, S., & Boyle, S. P. (2013). A specific, accurate, and sensitive measure of total plasma malondialdehyde by HPLC. *Journal of Lipid Research*, *54*(3), 852–858. https://doi.org/10.1194/jlr.D032698

Newton, G. L., Dorian, R., & Fahey, R. C. (1981). Analysis of biological thiols: Derivatization with monobromobimane and separation by reverse-phase high-performance liquid chromatography. *Analytical Biochemistry*, *114*(2), 383–387. https://doi.org/10.1016/0003-2697(81)

Pendyala, L., & Creaven, P. J. (1995). Pharmacokinetic and pharmacodynamic studies of N-acetylcysteine, a potential chemopreventive agent during a phase I trial. *Cancer, Epidemiology, Biomarkers & Prevention*, *4*(3), 245–251. https://pubmed.ncbi.nlm.nih.gov/7606199

Piknova, B., Park, J. W., Swanson, K. M., Dey, S., Noguchi, C. T., & Schechter, A. N. (2015). Skeletal muscle as an endogenous nitrate reservoir. *Nitric Oxide: Biology and Chemistry*, *47*, 10–16. https://doi.org/10.1016/j.niox.2015.02.145

Powers, S. K., & Jackson, M. J. (2008). Exercise-induced oxidative stress: Cellular mechanisms and impact on muscle force production. *Physiological Reviews*, *88*(4), 1243–1276. https://doi.org/10.1152/physrev.00031.2007

Reid, M. B., Haack, K. E., Franchek, K. M., Valberg, P. A., Kobzik, L., & West, M. S. (1992). Reactive oxygen in skeletal muscle I. Intracellular oxidant kinetics and fatigue in vitro. *Journal of Applied Physiology*, *73*(5), 1797–1804. https://doi.org/10.1152/jappl.1992.73.5.1797

Reid, M. B., Stokić, D. S., Koch, S. M., Khawli, F. A., & Leis, A. A. (1994). N-acetylcysteine inhibits muscle fatigue in humans. *Journal of Clinical Investigation*, *94*(6), 2468–2474. https://doi.org/10.1172/JCI117615

Senefeld, J. W., Wiggins, C. C., Regimbal, R. J., Dominelli, P. B., Baker, S. E., & Joyner, M. J. (2020). Ergogenic Effect of Nitrate Supplementation: A Systematic Review and Meta-analysis. *Medicine & Science in Sports & Exercise*, *52*(10), 2250–2261. https://doi.org/10.1249/MSS.0000000000002363

Shen, X., Kolluru, G. K., Yuan, S., & Kevil, C. G. (2015). Measurement of H2S in vivo and in vitro by the monobromobimane method. *Methods in Enzymology*, *554*, 31–45. https://doi.org/10.1016/bs.mie.2014.11.039

Sidhu, S. K., Cresswell, A. G., & Carroll, T. J. (2012). Motor cortex excitability does not increase during sustained cycling exercise to volitional exhaustion. *Journal of Applied Physiology (Bethesda, Md.: 1985)*, *113*(3), 401–409. https://doi.org/10.1152/japplphysiol.00486.2012

Tan, R., Wylie, L. J., Thompson, C., Blackwell, J. R., Bailey, S. J., Vanhatalo, A., & Jones, A. M. (2018). Beetroot juice ingestion during prolonged moderate-intensity exercise attenuates progressive rise in O_2 uptake. *Journal of Applied Physiology (Bethesda, Md.: 1985)*, *124*(5), 1254–1263. https://doi.org/10.1152/japplphysiol.01006.2017

Thurston, T. S., Weavil, J. C., Hureau, T. J., Gifford, J. R., Georgescu, V. P., Wan, H. Y., La Salle, D. T., Richardson, R. S., & Amann, M. (2021). On the implication of dietary nitrate supplementation for the hemodynamic and fatigue response to cycling exercise. *Journal of Applied Physiology (Bethesda, Md.: 1985)*, *131*(6), 1691–1700. https://doi.org/10.1152/japplphysiol.00400.2021

Wilkerson, D. P., Hayward, G. M., Bailey, S. J., Vanhatalo, A., Blackwell, J. R., & Jones, A. M. (2012). Influence of acute dietary nitrate supplementation on 50 mile time trial performance in well-trained cyclists. *European Journal of Applied Physiology*, *112*(12), 4127–4134. https://doi.org/10.1007/s00421-012-2397-6

Wootton-Beard, P. C., & Ryan, L. (2012). Combined use of multiple methodologies for the measurement of total antioxidant capacity in UK commercially available vegetable juices. *Plant Foods for Human Nutrition*, *67*(2), 142–147. https://doi.org/10/1007/s11130-012-0287-z

Wylie, L. J., Kelly, J., Bailey, S. J., Blackwell, J. R., Skiba, P. F., Winyard, P. G., Jeukendrup, A. E., Vanhatalo, A., & Jones, A. M. (2013a). Beetroot juice and exercise: Pharmacodynamic and dose-response relationships. *Journal of Applied Physiology (Bethesda, Md.: 1985)*, *115*(3), 325–336. https://doi.org/10.1152/japplphysiol.00372.2013

Wylie, L. J., Mohr, M., Krustrup, P., Jackman, S. R., Ermıdis, G., Kelly, J., Black, M. I., Bailey, S. J., Vanhatalo, A., & Jones, A. M. (2013b). Dietary nitrate supplementation improves team sport-specific intense intermittent exercise performance. *European Journal of Applied Physiology*, *113*(7), 1673–1684. https://doi.org/10.1007/s00421-013-2589-8